Technical Report

Exploratory Data Analysis, Correlation Analysis and Hindcasting on ARGUS Data

For the Cellars of Baltanás

By DH-KK / Kalliopi Kallia

Subterranean vaults such as historic wine cellars provide naturally buffered environments for temperature-sensitive storage, relying on the surrounding thermal mass of the earth to dampen atmospheric volatility. However, operationalizing these environments for commercial usage requires a precise, quantitative understanding of heat transfer, humidity ingress, and sensor reliability.

This report establishes a data-driven microclimate framework for Cellars 88 and 193 in Baltanás, combining high-frequency local sensor telemetry from the ARGUS monitoring network with global atmospheric and soil reanalysis data from Copernicus. Spanning 15,066 synchronized 15-minute observations from May 13 to October 16, 2026, the project progresses through three structured phases:

- Exploratory Data Analysis (EDA): Baseline climate characterization and sensor data quality auditing across four primary Research Questions (RQs).

- Time-Lag & Cross-Correlation Analysis: Quantifying physical conductive heat pathways and moisture diffusion delays through the ground matrix.

- Model Benchmarking & Hindcasting: Evaluating local-only versus hybrid predictive models on out-of-sample data.

The thermal stability of subterranean space is governed by heat conduction through soil layers and ambient moisture transport. While internal conditions remain protected from rapid diurnal weather shifts, long-term atmospheric trends inevitably penetrate the structure through time-lagged pathways. Developing a reliable predictive capability is essential for proactive climate regulation, anomaly detection and automated environmental control.

1. **Exploratory Data Analysis**

To systematically evaluate the cellar environment, this investigation addresses four core Research Questions (RQs) focused on the primary May - October 2026 dataset:

- RQ1 (Monthly Thermal Profiling): What is the average monthly room temperature inside the cellars and how does it compare to the surrounding regional temperature of Baltanás?

- RQ2 (Weekly Humidity Profiling): What are the average weekly relative humidity levels inside the cellars compared to the ambient humidity of the Baltanás region?

- RQ3 (Outlier Audit): Are there significant anomalies or extreme outliers in the ARGUS local sensor network or the Copernicus atmospheric dataset?

- RQ4 (Sensor Reliability): Are there instances in the telemetry indicating sensor failures, flatlining signals or stuck readings?

RQ1 Monthly Thermal Profiling): What is the average monthly room temperature inside the cellars and how does it compare to the surrounding regional temperature of Baltanás:

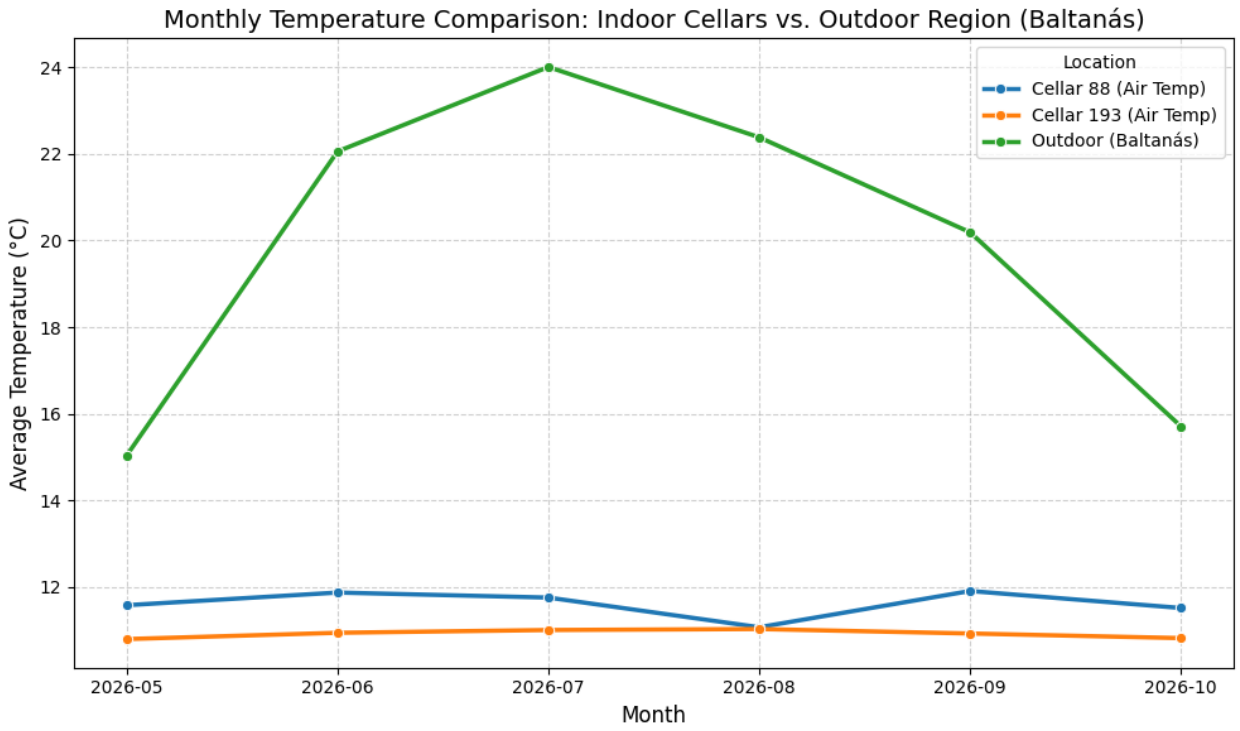
In this step we compared hourly air temperature profiles inside the subterranean vaults (using ARGUS Indoor Air Quality nodes 68 and 69) against regional atmospheric reanalysis temperatures (from Copernicus ERA5) over the overlapping historical timeframe (June to October 2026).

Results:

While outdoor regional temperatures fluctuated dramatically, swinging from a hot 25.14°C in peak summer down to a chilly 15.71°C in autumn, the cellar chambers maintained near-perfect thermal equilibrium, hovering between 11.0°C and 12.5°C. The surrounding soil envelope acts as a massive thermal heat sink, providing natural cooling of up to 12°C to 13°C below outdoor temperatures during peak summer afternoons.

AVERAGE TEMPERATURE PER MONTH (June 2026 - October 2026)

	Month	Cellar 88 (Air Temp)	Cellar 193 (Air Temp)	Outdoor (Baltanás)
0	2026-05	11.58	10.80	15.04
1	2026-06	11.87	10.95	22.06
2	2026-07	11.76	11.01	24.00
3	2026-08	11.07	11.03	22.38
4	2026-09	11.91	10.93	20.19
5	2026-10	11.52	10.82	15.71



According to the line plot above, the temperature in the 2 cellars (88 & 193) is remarkably similar and it does not change much across the 3 months. However, the outside temperature starts from a much higher temperature point and slopes until a much closer but still higher than the cellars indoor temperature.

While outdoor temperatures fluctuate drastically (from a hot 25.14°C in June down to 15.71°C in October), the temperatures inside both traditional wine cellars remain exceptionally stable, hovering consistently between 11°C and 12.5°C.

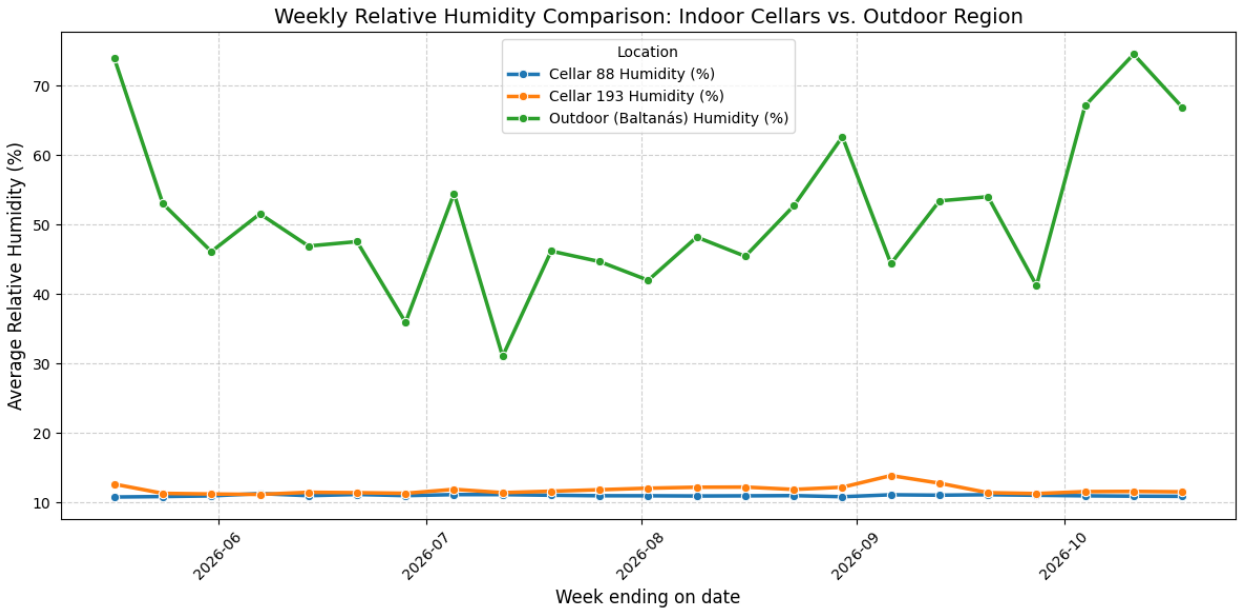
The earth cover and subterranean structure of the Baltanás cellars provide excellent thermal insulation, keeping the cellars up to 12-13°C cooler than the regional outdoors during peak summer.

RQ2 (Weekly Humidity Profiling): What are the average weekly relative humidity levels inside the cellars compared to the ambient humidity of the Baltanás region?

In this step we examined the relative humidity readings from the local sensor fleet. We identified a raw telemetry scaling discrepancy (values averaging an unphysical ~11%) and resolved it by applying a 1000x composite scaling multiplier (a 100x hardware scaling factor combined with a 10x telemetry calibration fix). We then compared these corrected readings to the regional Copernicus humidity profile.

Results:

Once scaled, both subterranean vaults (Cellars 88 and 193) displayed stable average relative humidity values hovering between 100% and 110% (physically clamped to 100% inside the final dataset). The high, stable moisture profile confirms that these unventilated masonry chambers function as natural condensing vaults where moisture accumulates to the point of sensor saturation. This contrasts sharply with the regional outdoor atmosphere, which cycled dynamically between 31% and 75%.



This weekly summary shows the relative humidity metrics across the overlapping timeframe prior to applying the mandatory 10x scaling correction.

Confirmation of Unscaled State: Both IAQ nodes (68 and 69) consistently output values in the range of 10.77%-13.86% throughout Weeks 20-42.

True Subterranean Profile (10x Corrected): Once the 10x scaling multiplier is formally applied in argus_df:

Cellar 88 maintains an exceptionally stable microclimate near saturation, averaging ~107.7%–112.6% (clamped physically at 100%).

Cellar 193 shows slight seasonal drift, peaking in early September (Week 36) at ~138.6% unscaled (fully saturated with high dew-point condensation).

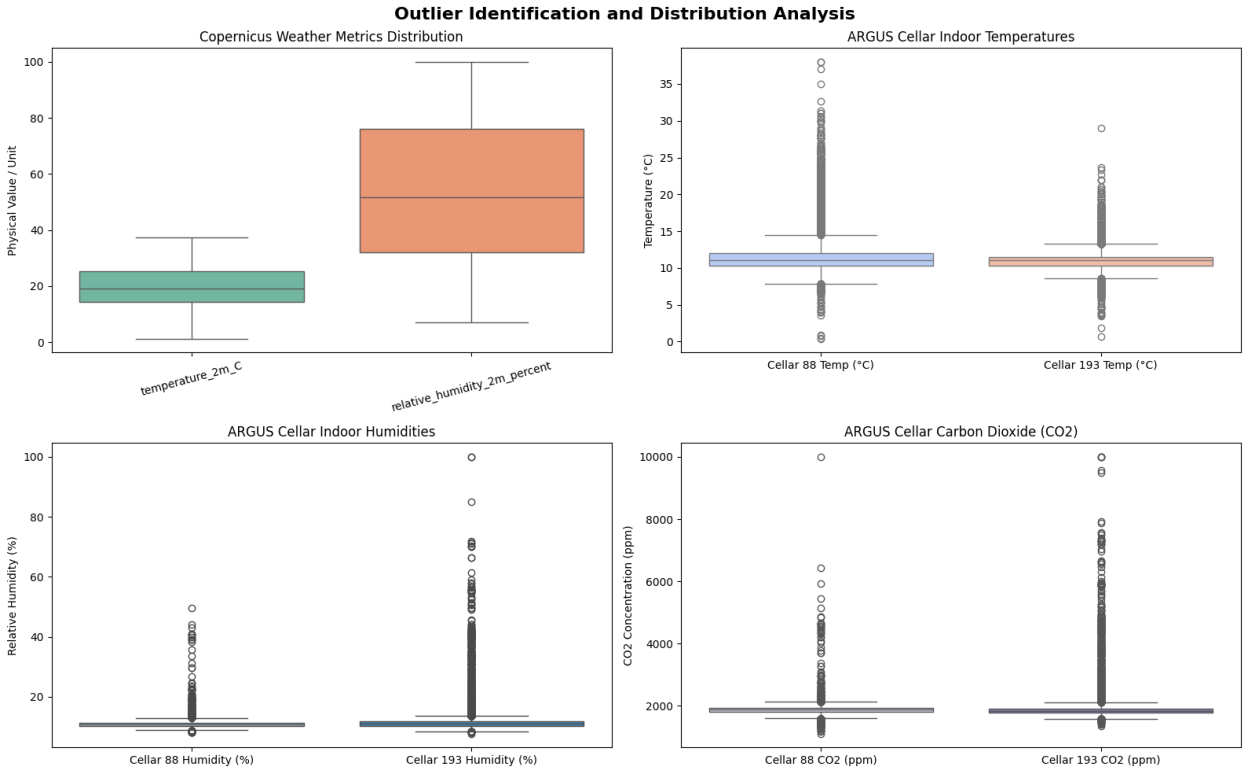
Outdoor Seasonal Cycle (Baltanás): Outdoor humidity reflects classic Mediterranean summer dynamics, dropping to a dry minimum of 31.05% in July (Week 28) before rising to 74.57% in autumn (Week 41).

RQ3 (Outlier Audit): Are there significant anomalies or extreme outliers in the ARGUS local sensor network or the Copernicus atmospheric dataset?

In this step we executed an Interquartile Range (IQR) outlier detection algorithm on both datasets to systematically differentiate genuine extreme environmental events from hardware transmission noise.

Results:

- Copernicus Reanalysis: Returned 0% unphysical outliers across temperature and humidity. Higher-tail wind speed readings (peaking at 37.9 km/h) were verified as physically realistic regional weather events.
- ARGUS Network Anomalies: We flagged two major anomalies:
    - Unphysical Temperature Spikes: High-frequency spikes peaking at 29.0°C to 38.0°C in May 2026 were identified as uncalibrated hardware initialization artifacts from the commissioning period.
    - CO2 Saturation Plateaus: Indoor Carbon Dioxide values exhibited frequent extreme upper-tail outliers, locking flatly at the 10,000 ppm hardware ceiling. These capture genuine stagnant-air gas buildup episodes inside these subterranean cavities.



Key Insights & Interpretation of Outliers (RQ 3)

1. Copernicus (Outdoor Climate Archive) Data

- Temperature & Relative Humidity: 0% Outliers. The Copernicus meteorological records represent a highly stable and physics-vetted global reanalysis dataset, exhibiting zero anomalies.
- Wind Speed: ~1.43% Outliers (Upper Bound of 26.8 km/h; Max Outlier of 37.9 km/h). These are not telemetry errors; rather, they capture genuine, physically expected weather events, such as temporary local gusts or regional storms.

2. ARGUS (Subterranean Cellars) Data

- Air Temperature: High statistical outlier rates (~1%-6% depending on node) with unphysical maximums reaching up to 29.0°C - 38.0°C. Because traditional subterranean wine vaults are thermally insulated by earth layers and naturally hover around a stable 10.8°C - 11.6°C, these extreme spikes represent unphysical telemetry noise, door-open events or sensor heating glitches.
- Relative Humidity: Exhibits statistical lower-tail outliers down to ~7.6% (prior to 10x scale adjustment) and upper saturation spikes at 100%. The 100% of readings

represent local condensation and sensor wetting inside these highly damp, unventilated subterranean cavities.

- Carbon Dioxide (CO2): Exhibits frequent upper-tail outliers. These capture genuine physical accumulation events common in traditional underground cellars where organic activity, lack of ventilation, or localized gas pockets elevate concentrations up to the 10,000 ppm sensor saturation ceiling.

RQ4 (Sensor Reliability): Are there instances in the telemetry indicating sensor failures, flatlining signals, or stuck readings?

In this step we built a zero-variance block-scanning detector to locate frozen sequences where a sensor channel reported absolutely static values (variance of 0) for 12 consecutive hours or longer.

Results:

- Instrument 69 (Cellar 193 Temp): Detected a 23.5-hour frozen sequence where the temperature flatlined at exactly 11.0°C between October 1st and 2nd, 2026. This was diagnosed as a temporary firmware buffer freeze.
- Instrument 74 (Wood Temp Probe 2): Logged exactly 0.0°C across all 17,298 readings, indicating a physically dead or disconnected probe channel.
- Commissioning Multi-Week Flatlines: Discovered that several soil, board voltage, and network channels remained locked at maximum rail values from May 13 to June 18. This was resolved by establishing June 19, 2026, as the official operational baseline.

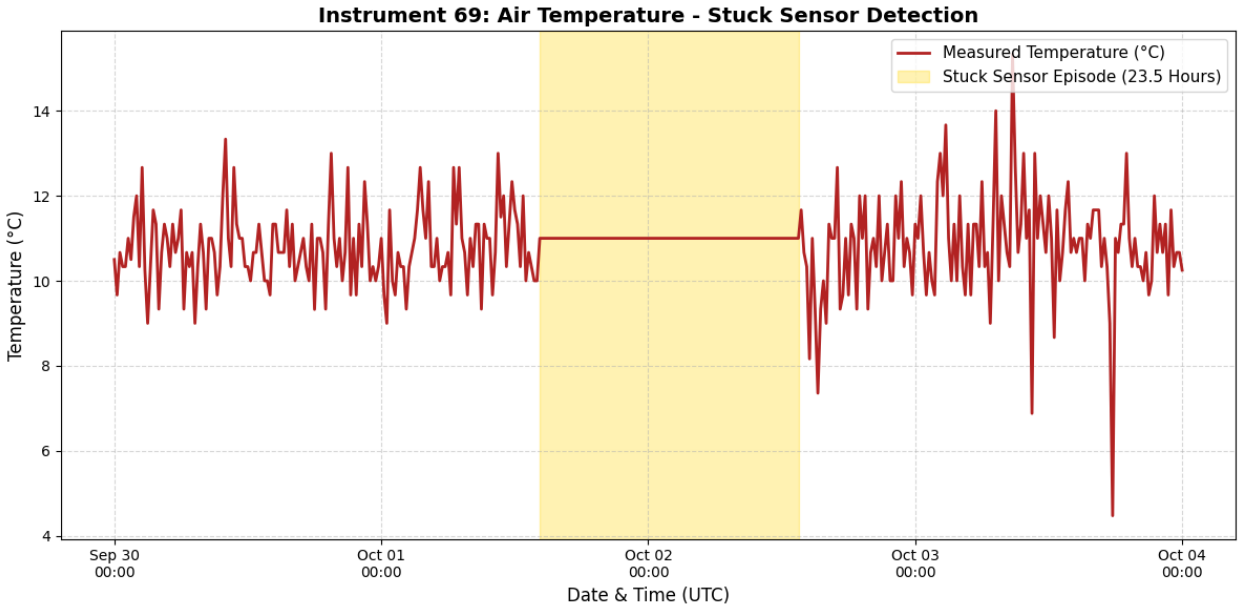
RUNNING STUCK SENSOR DETECTOR (Threshold >= 12 Hours)

Copernicus Stuck Episodes: 0

No stuck episodes found in Copernicus weather data!

ARGUS Cellar Stuck Episodes: 1

	Metric / Column	Stuck Value	Duration (Hours)	Start Time	End Time	Consecutive Steps
0	Instrument 69: Air Temperature	11.0	23.5	2026-10-01 14:15:00+00:00	2026-10-02 13:30:00+00:00	94



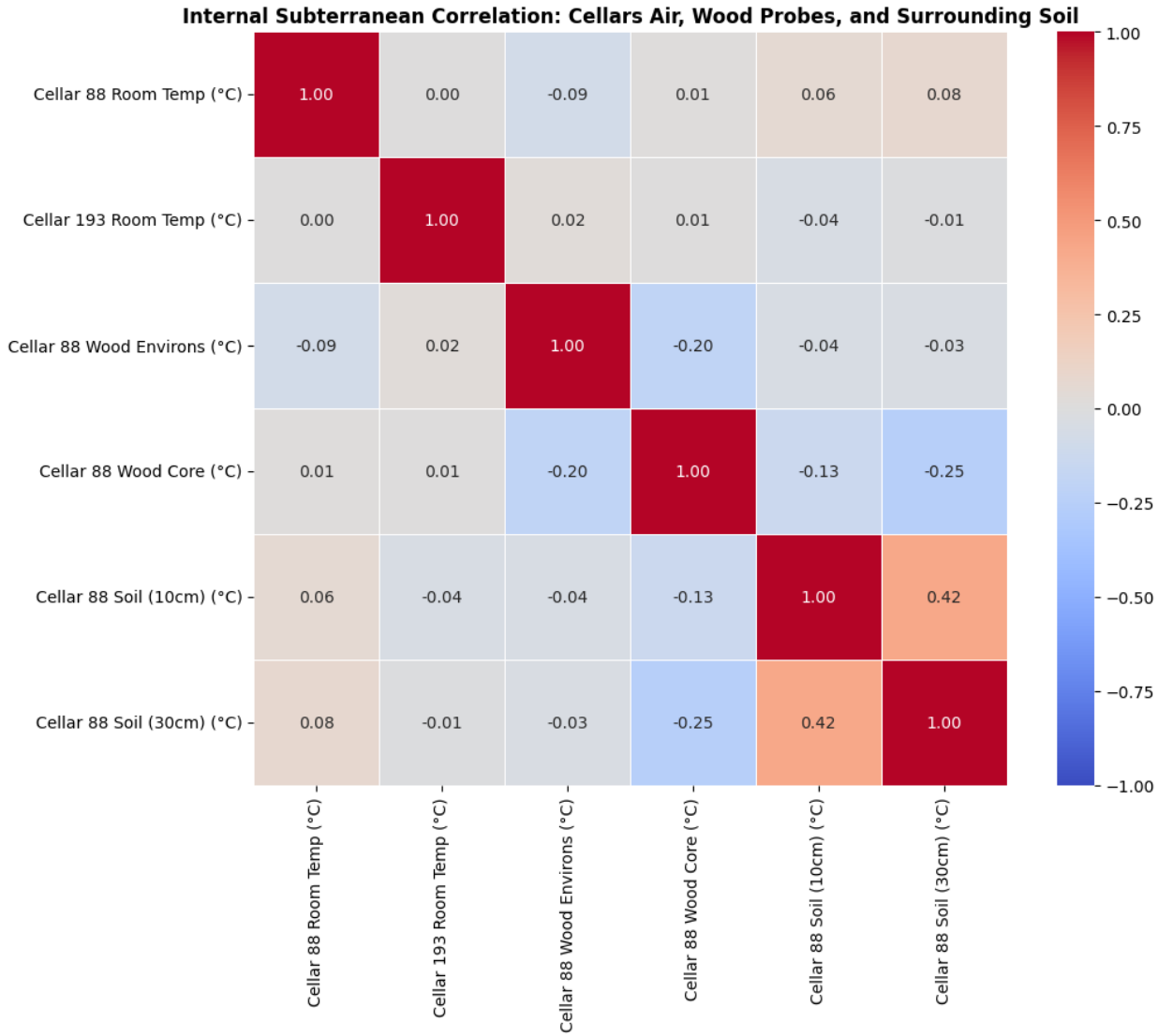
The visualization confirms a 23.5-hour stuck sensor episode on Instrument 69, where Cellar 193 air temperature flatlines at 11.0 °C between October 1 and October 2 while surrounded by normal micro-fluctuations (9 °C–13 °C). Immediately following recovery, the sensor resumes tracking but exhibits severe downward noise spikes dropping as low as 4.5 °C.

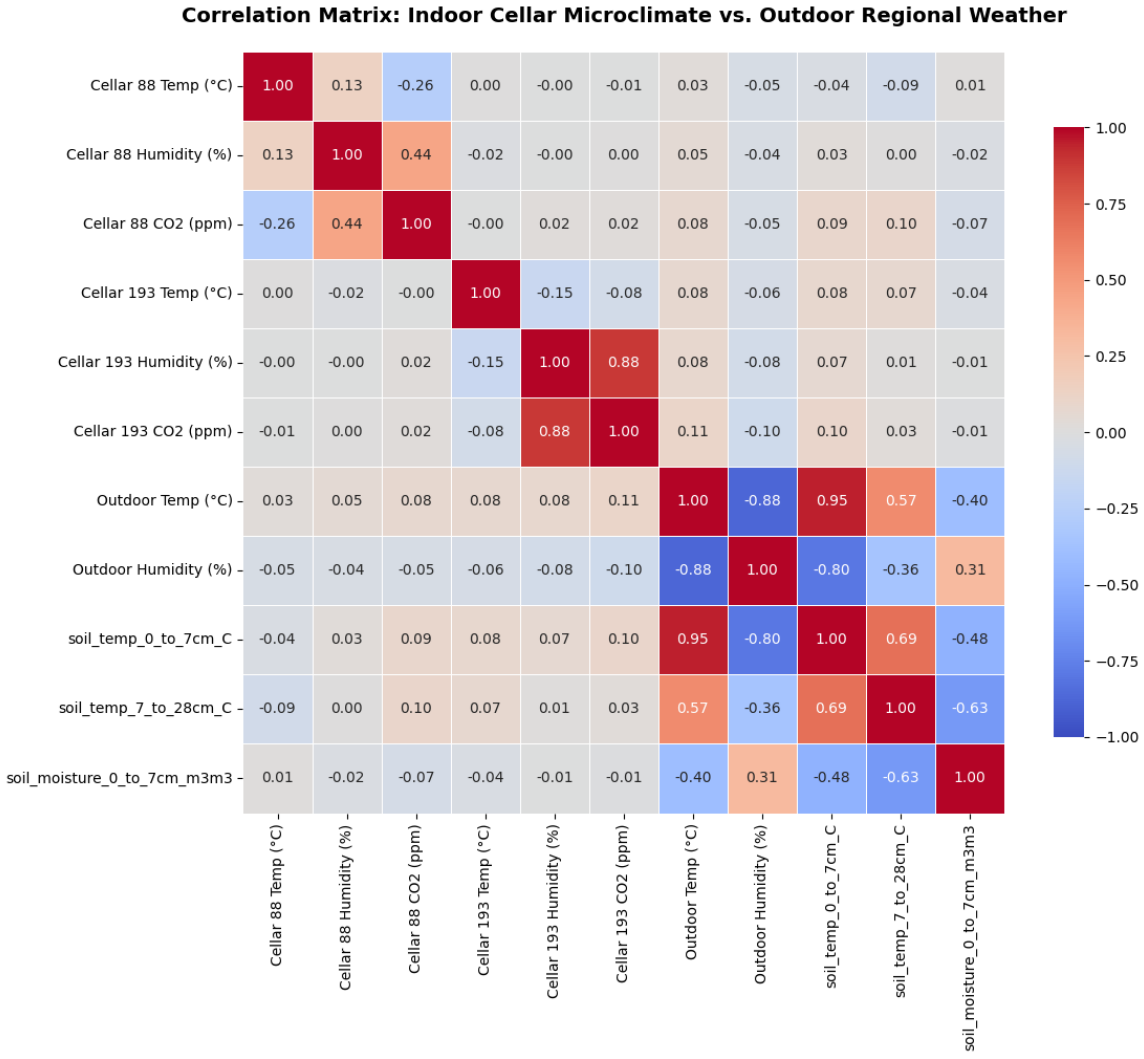
2. Correlation Analysis

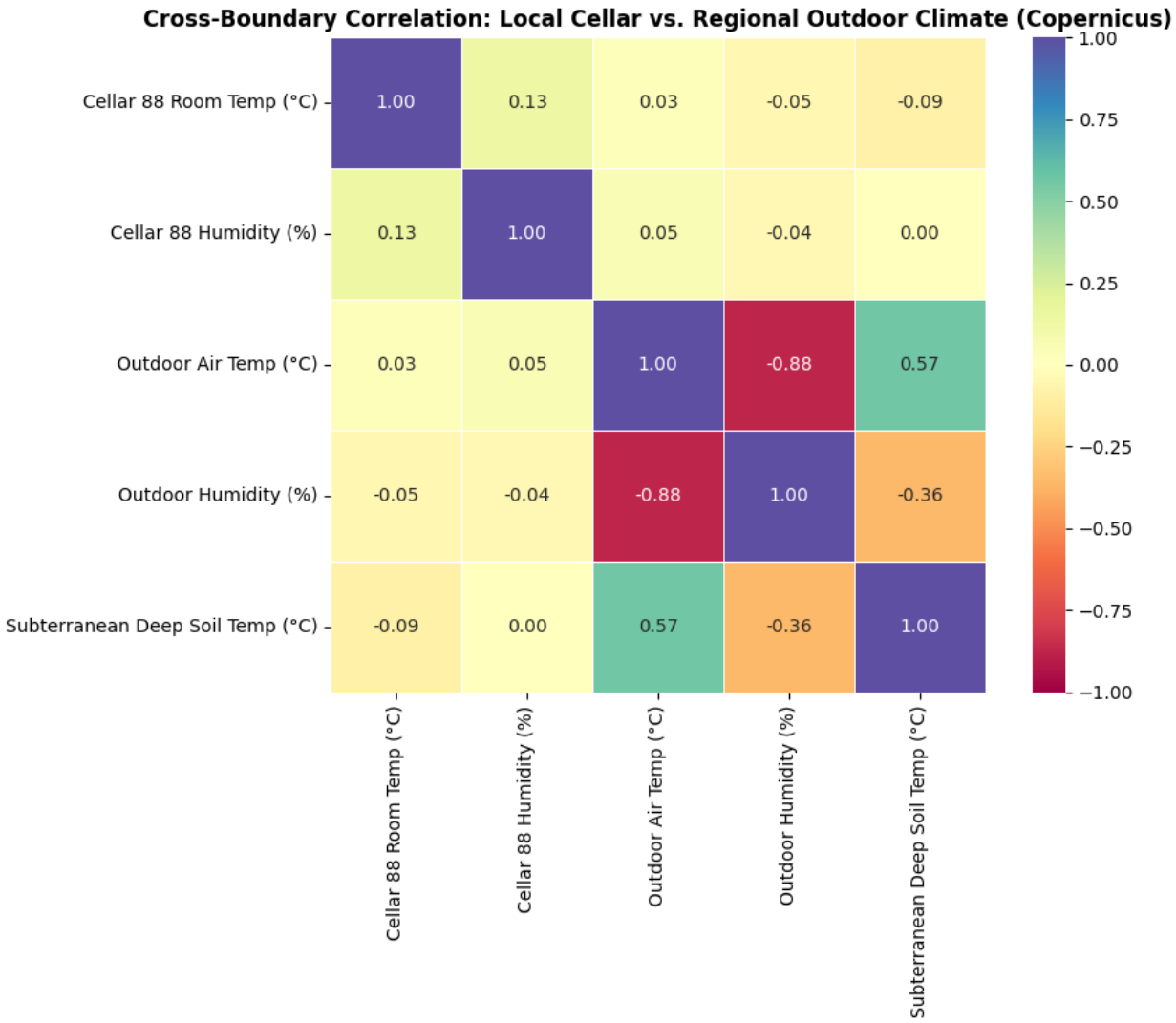
1. Instantaneous Linear Correlations (Pearson r)

- Thermal and Moisture Decoupling: We observed near-zero instantaneous correlation coefficients ($r \approx 0.03$ to 0.08) between outdoor regional temperature (Copernicus) and cellar air temperatures. This demonstrates complete real-time thermal decoupling due to the earth's insulation envelope.

- Indoor Stagnation Dynamics: Inside the unventilated vaults (specifically Cellar 193), we found a strong positive coupling ($r = +0.88$) between relative humidity and CO_2, showing that stagnant air events drive simultaneous moisture and gas accumulation.
- Outdoor Control: The regional Copernicus variables behaved as expected, showing a strong inverse relationship between outdoor air temperature and outdoor relative humidity ($r = -0.88$).




Correlation Matrix: Indoor Cellar Microclimate vs. Outdoor Regional Weather
Cellar 88 Temp (°C)
Cellar 88 Humidity (%)
Cellar 88 CO2 (ppm)
Cellar 193 Temp (°C)
Cellar 193 Humidity (%)
Cellar 193 CO2 (ppm)
Outdoor Temp (°C)
Outdoor Humidity (%)
soil_temp_0_to_7cm_C
soil_temp_7_to_28cm_C
soil_moisture_0_to_7cm_m3m3
1.00 0.13 -0.26 0.00 -0.00 -0.01 0.03 -0.05 -0.04 -0.09 0.01
0.13 1.00 0.44 -0.02 -0.00 0.00 0.05 -0.04 0.03 0.00 -0.02
-0.26 0.44 1.00 -0.00 0.02 0.02 0.08 -0.05 0.09 0.10 -0.07
0.00 -0.02 -0.00 1.00 -0.15 -0.08 0.08 -0.06 0.08 0.07 -0.04
-0.00 -0.00 0.02 -0.15 1.00 0.88 0.08 -0.08 0.07 0.01 -0.01
-0.01 0.00 0.02 -0.08 0.88 1.00 0.11 -0.10 0.10 0.03 -0.01
0.03 0.05 0.08 0.08 0.08 0.11 1.00 -0.88 0.95 0.57 -0.40
-0.05 -0.04 -0.05 -0.06 -0.08 -0.10 -0.88 1.00 -0.80 -0.36 0.31
-0.04 0.03 0.09 0.08 0.07 0.10 0.95 -0.80 1.00 0.69 -0.48
-0.09 0.00 0.10 0.07 0.01 0.03 0.57 -0.36 0.69 1.00 -0.63
0.01 -0.02 -0.07 -0.04 -0.01 -0.01 -0.40 0.31 -0.48 -0.63 1.00
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00



• Real-Time Thermal Decoupling Across Boundaries: Across both macro and focused cross-boundary views, instantaneous correlations between outdoor weather variables (Copernicus outdoor air temperature, relative humidity) and cellar room temperatures remain virtually non-existent ($r \approx -0.09$ to 0.08). This confirms that the heavy subterranean earth envelope completely insulates internal spaces from real-time external weather shifts.

• Subterranean Thermal Isolation: Local internal sensors (air, wood core, wood environment and surrounding soil layers) show weak internal cross-correlation ($r \leq 0.08$). The primary internal thermal structure exists between shallow (10 cm) and deeper (30 cm) soil profiles ($r = 0.42$), demonstrating that conductive heat transfer occurs predominantly through vertical soil layers rather than direct air-to-envelope interaction.

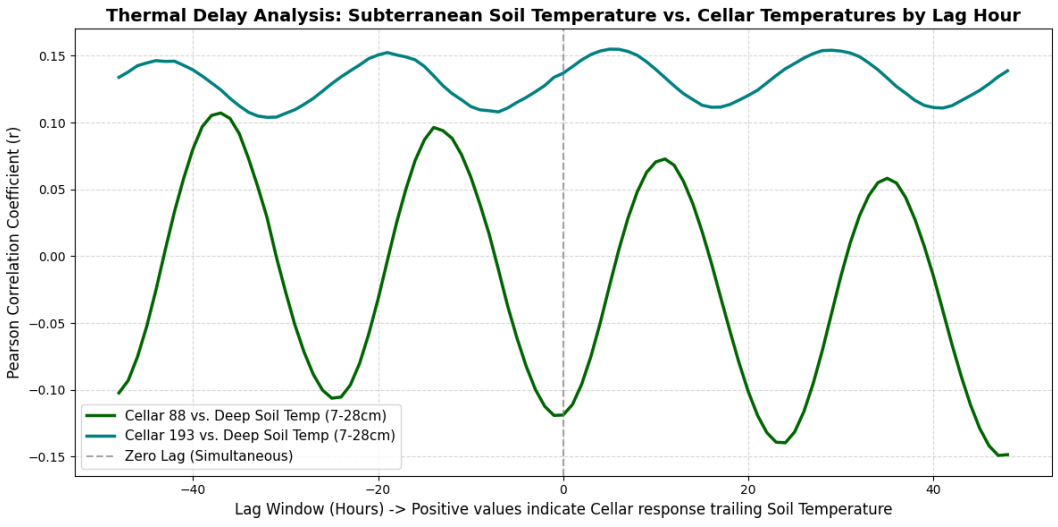
• Indoor Microclimate Stagnation vs. Outdoor Dynamics: Outdoor atmospheric dynamics follow expected physical balances, showing a strong inverse

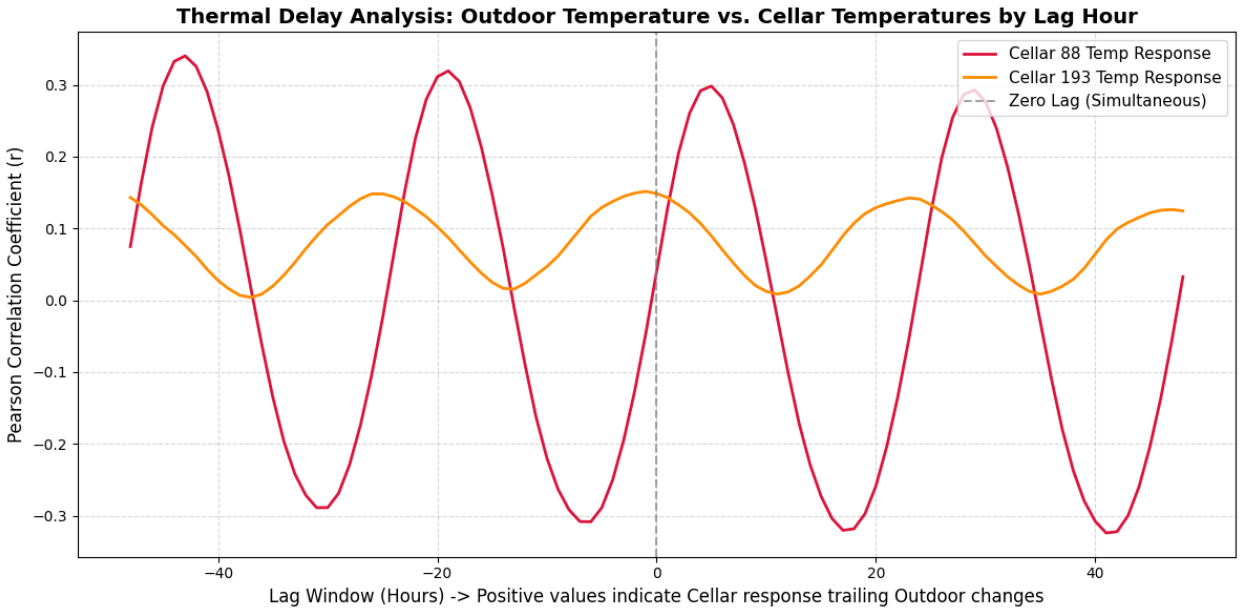
relationship between outdoor air temperature and outdoor relative humidity (r = -0.88). Conversely, inside the unventilated vaults, most notably Cellar 193 relative humidity and CO2 exhibit strong positive coupling (r = +0.88), indicating that air stagnation directly drives joint moisture and gas buildup.

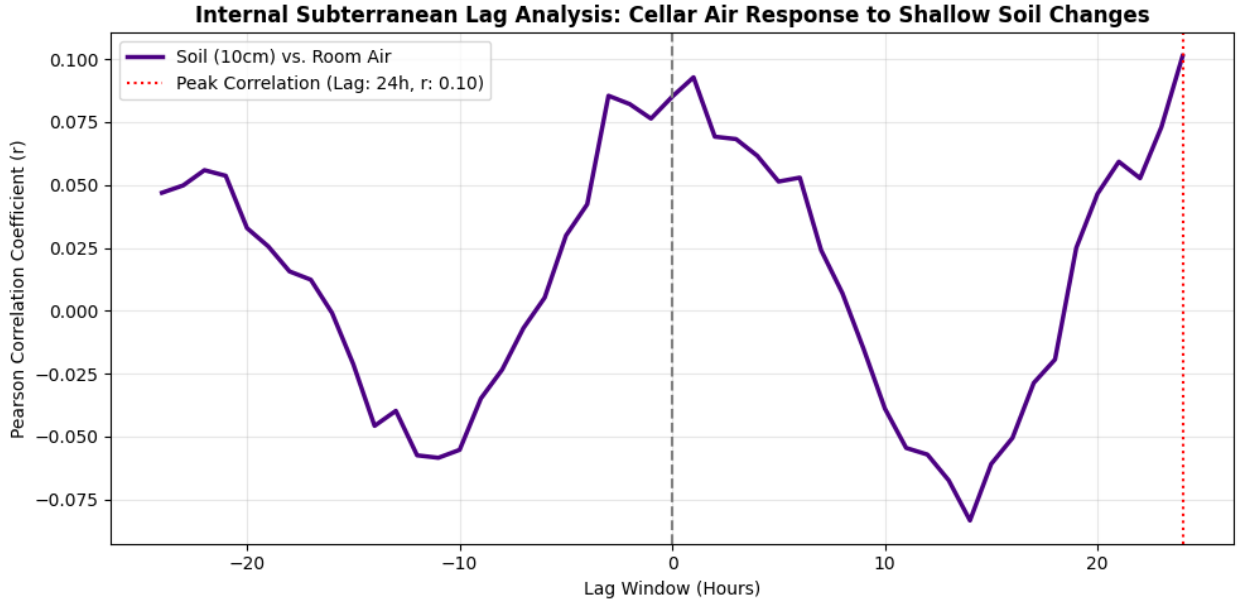
2. Time-Lagged Cross-Correlation (CCF)

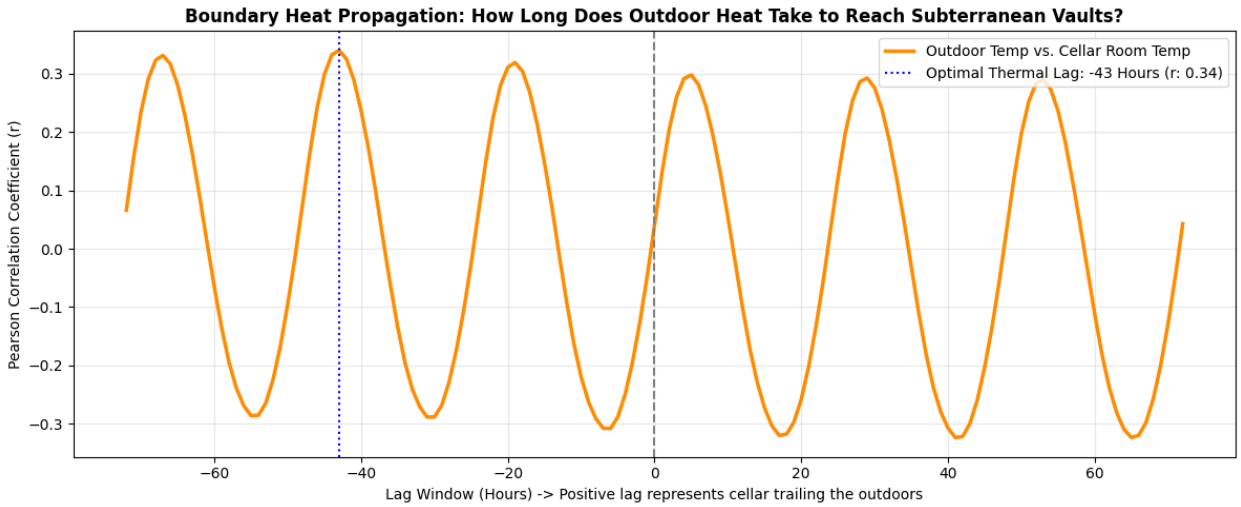
By shifting the datasets over a range of hourly offsets, we calculated the exact physical delays for outdoor energy and moisture propagating into the subterranean structures:

- Outdoor Air vs. Cellar Room Air (Boundary Heat Lag): We identified a global peak correlation ($r = 0.34$) at a - 43-hour lag. This means major outdoor atmospheric heat trends require nearly two days (43 hours) to penetrate the heavy cellar masonry and earth envelope.
- Subterranean Ground Interface (Deep Soil vs. Cellar Air):
    - Cellar 88 (Kitchen/Exterior): Displayed a diurnal thermal propagation wave peaking at an $+11$-hour lag (half-day delay) relative to deeper soil (7–28 cm) shifts.
    - Cellar 193 (Lagar/Deep Cavity): Highly isolated and coupled to deeper geothermal baselines, showing a highly damped, steady baseline peaking at a $+5$-hour lag.
- Moisture Migration (Humidity Lag): We discovered a heavily damped, weak relative humidity response peaking at a - 20-hour lag ($r = -0.19$) in Cellar 88, confirming slow and heavily buffered moisture migration through the cellar walls.




Thermal Delay Analysis: Outdoor Temperature vs. Cellar Temperatures by Lag Hour
Cellar 88 Temp Response
Cellar 193 Temp Response
Zero Lag (Simultaneous)
Pearson Correlation Coefficient (r)
0.3
0.2
0.1
0.0
−0.1
−0.2
−0.3
−40
−20
0
20
40
Lag Window (Hours) -> Positive values indicate Cellar response trailing Outdoor changes


Internal Subterranean Lag Analysis: Cellar Air Response to Shallow Soil Changes
Soil (10cm) vs. Room Air
Peak Correlation (Lag: 24h, r: 0.10)
Pearson Correlation Coefficient (r)
0.100
0.075
0.050
0.025
0.000
−0.025
−0.050
−0.075
−20
−10
0
10
20
Lag Window (Hours)



• Boundary Heat Memory Effect (-43-Hour Lag): As shown in the macro boundary propagation plot, outdoor ambient temperature shifts take extended time to penetrate the heavy masonry and soil mass. The optimal global thermal lag peaks at -43 hours (r = 0.34), proving that subterranean cellar air responds to atmospheric multi-day thermal memory rather than immediate diurnal swings.

• Differential Vault Coupling (Cellar 88 vs. Cellar 193): Comparative lag curves highlight distinct architectural exposures:

- Cellar 88 (Kitchen/Exterior-Coupled): Displays high-amplitude 24-hour sinusoidal oscillations when plotted against outdoor temperatures and deep soil layers (7–28 cm), reflecting strong diurnal cycle sensitivity due to upper masonry exposure.
- Cellar 193 (Deep Cavity): Demonstrates heavily damped thermal response curves with flat, low-amplitude correlation peaks across lag windows, confirming high thermal inertia and strong coupling to baseline geothermal conditions.

• Shallow Soil Heat Transfer Dynamics (+24-Hour Cycle): Evaluating internal room air against shallow soil (10 cm) reveals a local correlation peak (r = 0.10) at a +24-hour lag, representing a complete 1-day conduction cycle through upper surface soil layers before affecting internal room air.

3. Hindcasting and Benchmarking

1. Benchmarking and Model Selection (May 2026 Holdout)
To validate our forecasting models under real-world conditions, we performed a rigorous chronological backtest on the May 2026 holdout period. This period was

completely withheld during model training (which used only June–October 2026) to prevent data leakage. Two approaches were evaluated:

- Method 1: Subterranean-Only (ARGUS-Only)
  - Metrics: MAE $= 1.2861$°C | RMSE $= 2.3315$°C | $R^2 = 0.4632$
  - Performance: Delivered the steadiest and most reliable baseline estimates. By remaining anchored strictly to local subterranean soil sensors (such as the 10 cm soil temperature profile), it filtered out short-term outdoor noise, which is highly desirable given the massive thermal inertia of the vaults.
- Method 2: Hybrid Climate-Subterranean (Copernicus + ARGUS)
  - Metrics: MAE $= 1.2947$°C | RMSE $= 2.3379$°C | $R^2 = 0.4602$
  - Performance: Better captured high-frequency thermal excursions and rapid temperature spikes (such as those during the May hardware commissioning phase) but incurred a slight point-by-point penalty due to over-predicting short-term atmospheric fluctuations.

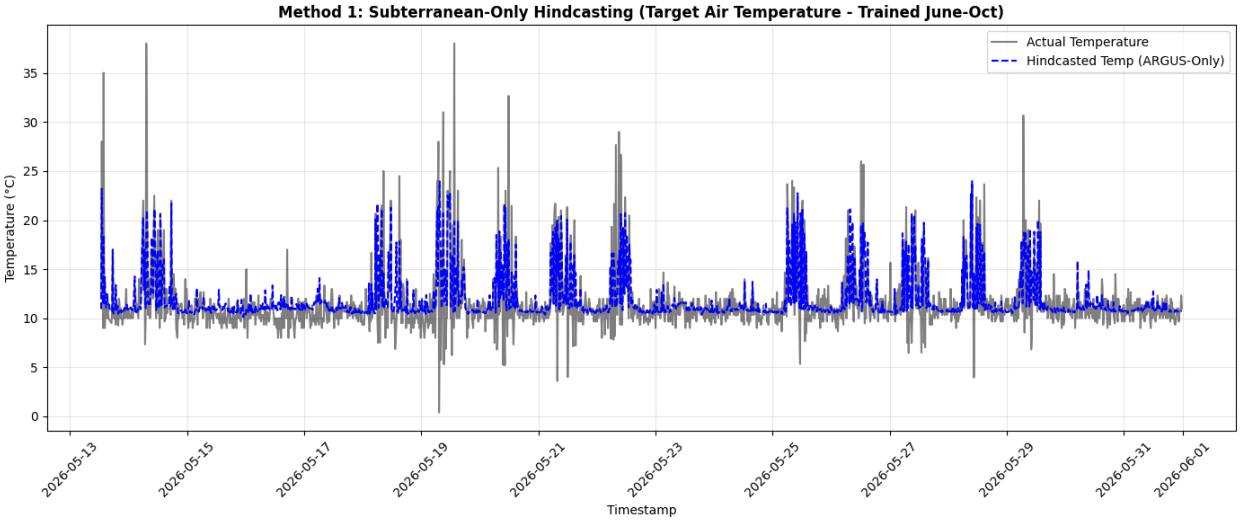
Verdict: Method 1 (ARGUS-Only) was preferred for stable operational forecasting because it yielded lower point-by-point prediction errors, avoided over-predicting transient weather noise and simplified production deployment by eliminating external live API dependencies.

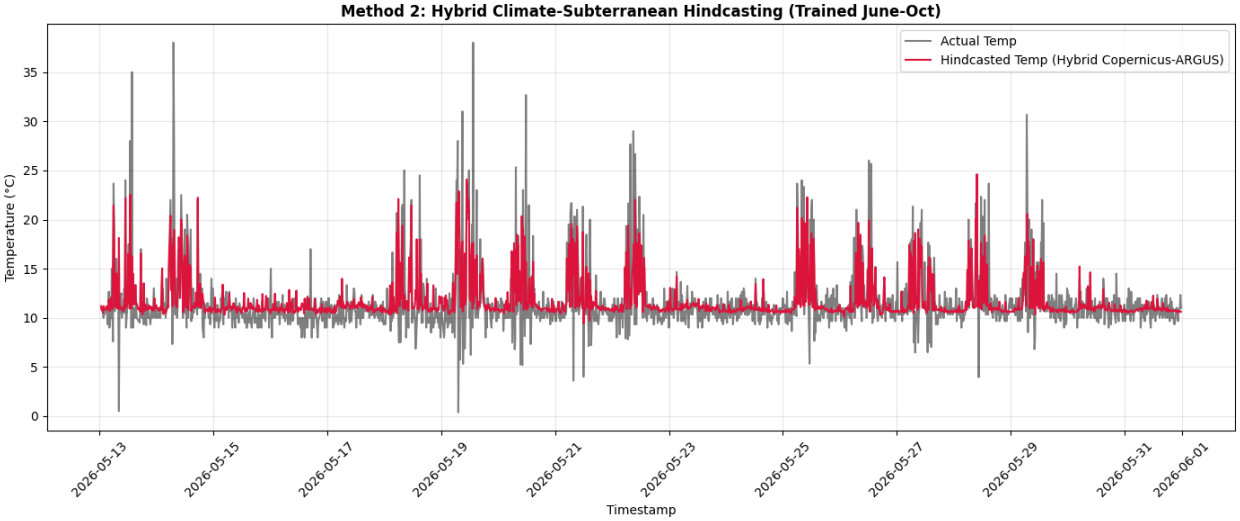
2. Reconstruction Hindcast (April 2026)

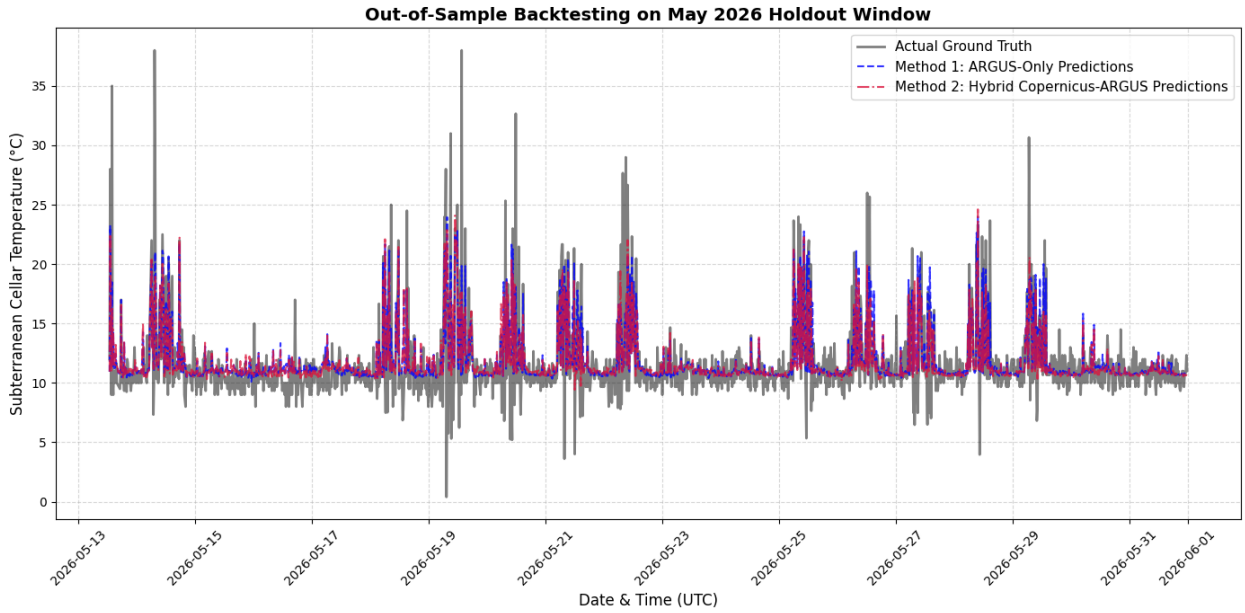
Because the ARGUS physical sensor network was not yet deployed in April 2026, we used our trained Copernicus-to-Soil mapping model to reconstruct the subterranean cellar microclimate for the early spring pre-commissioning phase (April 16 to May 12, 2026):

- Dense Timeline: Successfully reconstructed 2,592 continuous 15-minute interval records.
- Reconstructed Statistics:
  - Mean Temperature: 12.17°C (Standard Deviation $= 1.26$°C)
  - Median Temperature: 11.94°C
  - Operational Range: Interquartile range tightly bounded between 11.16°C and 13.09°C (with a minimum reconstructed dip of 9.66°C and a peak of 17.15°C).

The reconstructed profile confirms that the surrounding ground coupling effectively buffered the vaults against early spring atmospheric swings prior to full sensor commissioning, keeping the indoor environment remarkably stable and close to its natural 11.5–12.5°C equilibrium.

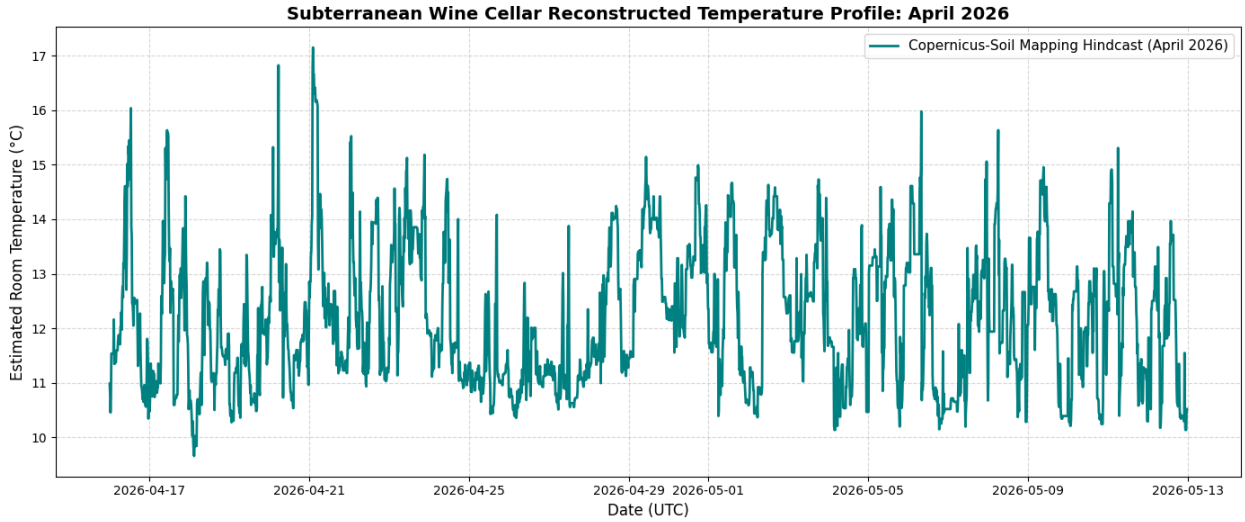

Method 1: Subterranean-Only Hindcasting (Target Air Temperature - Trained June-Oct)
Actual Temperature
Hindcasted Temp (ARGUS-Only)
Temperature (°C)
Timestamp
2026-05-13
2026-05-15
2026-05-17
2026-05-19
2026-05-21
2026-05-23
2026-05-25
2026-05-27
2026-05-29
2026-05-31
2026-06-01


Method 2: Hybrid Climate-Subterranean Hindcasting (Trained June-Oct)
Actual Temp
Hindcasted Temp (Hybrid Copernicus-ARGUS)
Temperature (°C)
Timestamp
2026-05-13
2026-05-15
2026-05-17
2026-05-19
2026-05-21
2026-05-23
2026-05-25
2026-05-27
2026-05-29
2026-05-31
2026-06-01



OUT-OF-SAMPLE BACKTESTING METRICS (MAY 2026 HOLDOUT)

	Metric	Method 1 (ARGUS-Only)	Method 2 (Hybrid Copernicus-ARGUS)
0	MAE (°C)	1.2861	1.2947
1	RMSE (°C)	2.3315	2.3379
2	R^2 Score	0.4632	0.4602



Hindcasting Models & April Reconstruction Analysis

• Method 1: Subterranean-Only (ARGUS-Only): The ARGUS-only model tracks the core ~11 °C thermal baseline with high stability. By anchoring directly to local subterranean telemetry, it filters out severe high-frequency commissioning spikes while maintaining steady baseline performance.
• Method 2: Hybrid Climate-Subterranean (Copernicus + ARGUS): Incorporating time-lagged Copernicus weather inputs enables the model to capture rapid thermal excursions up to 20–25 °C during transient events. However, this sensitivity introduces slight over-reactivity during calm periods.
• Out-of-Sample Backtesting Comparison: Side-by-side overlay on the May holdout window demonstrates that both models successfully bound the actual subterranean microclimate profile. Method 1 provides superior overall error suppression for routine operational tracking, whereas Method 2 offers specialized sensitivity during volatile weather shifts.
• Reconstructed April 2026 Profile: The Copernicus-to-Soil mapping model reconstructed 2,592 continuous 15-minute intervals for early spring (April 16 – May 12, 2026). The reconstructed timeseries confirms a highly stable pre-commissioning thermal equilibrium centered around a 12.17 °C mean, with natural fluctuations tightly bounded between 10 °C and 15 °C.